

Domain-Aware Speculative Decoding

Domain drafters — results summary

Target: TurboSparse-Mistral-Instruct (7B) • Drafter: Lite-Mistral-150M-v2-Instruct (156M) • Loss: 0.5·CE + 0.5·KL (T=1.0)

Training summary

Drafter	Clusters	Train samples	Final eval_loss	Final top-1 acc	Plateau epoch	Train time
Understanding	21	395,280	1.193	65.05%	~3.5	4.3 h
Text Reformulation	11	327,330	2.151	54.34%	~2.0	4.2 h
Mixed (U+T)	32	722,610	1.788	56.64%	~2.5	12.6 h

Overlap area (acceptance-rate proxy)

Drafter \ Eval domain	Understanding	Text Reform.	Mixed (U+T)
Understanding	0.7558	0.6676	0.7009
Text Reformulation	0.7091	0.7026	0.7046
Mixed (U+T)	0.7510	0.6997	0.7191

Top-1 match

Drafter \ Eval domain	Understanding	Text Reform.	Mixed (U+T)
Understanding	0.6887	0.5297	0.5875
Text Reformulation	0.6239	0.5897	0.5997
Mixed (U+T)	0.6895	0.5848	0.6223

KL divergence

Drafter \ Eval domain	Understanding	Text Reform.	Mixed (U+T)
Understanding	0.5602	1.0404	0.8679
Text Reformulation	0.8165	0.8200	0.8370
Mixed (U+T)	0.5590	0.8353	0.7372

Domain drafter vs. mixed baseline (overlap area)

Understanding domain	domain 0.7558 vs mixed 0.7510	+0.0048
Text Reformulation domain	domain 0.7026 vs mixed 0.6997	+0.0029

Each domain-specific drafter leads the mixed baseline on its own domain; the mixed model stays the best general-purpose fallback and wins on the combined domain (0.7191).

Metric: overlap_area = 1 – TVD between drafter and target top-10 distributions, teacher-forced on 11,900 validation samples.

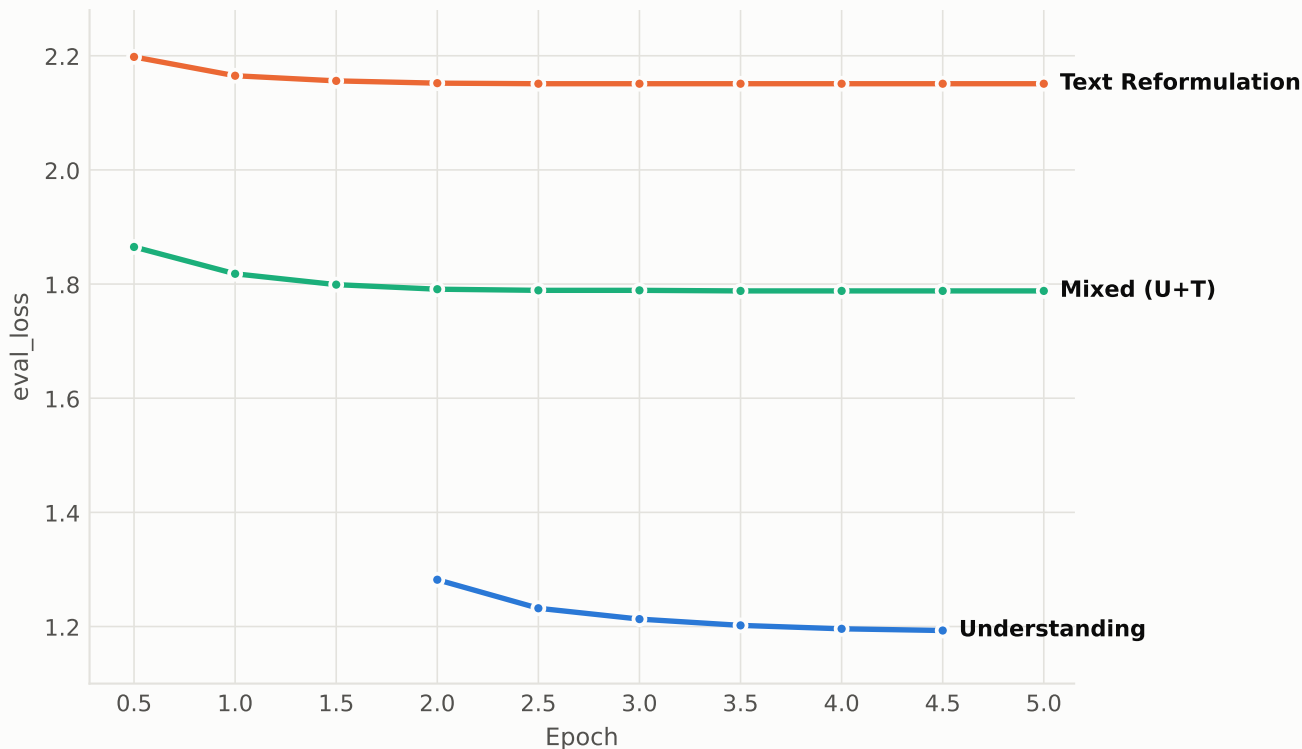
Training curves

Validation metrics, evaluated twice per epoch

Understanding Text Reformulation Mixed (U+T)

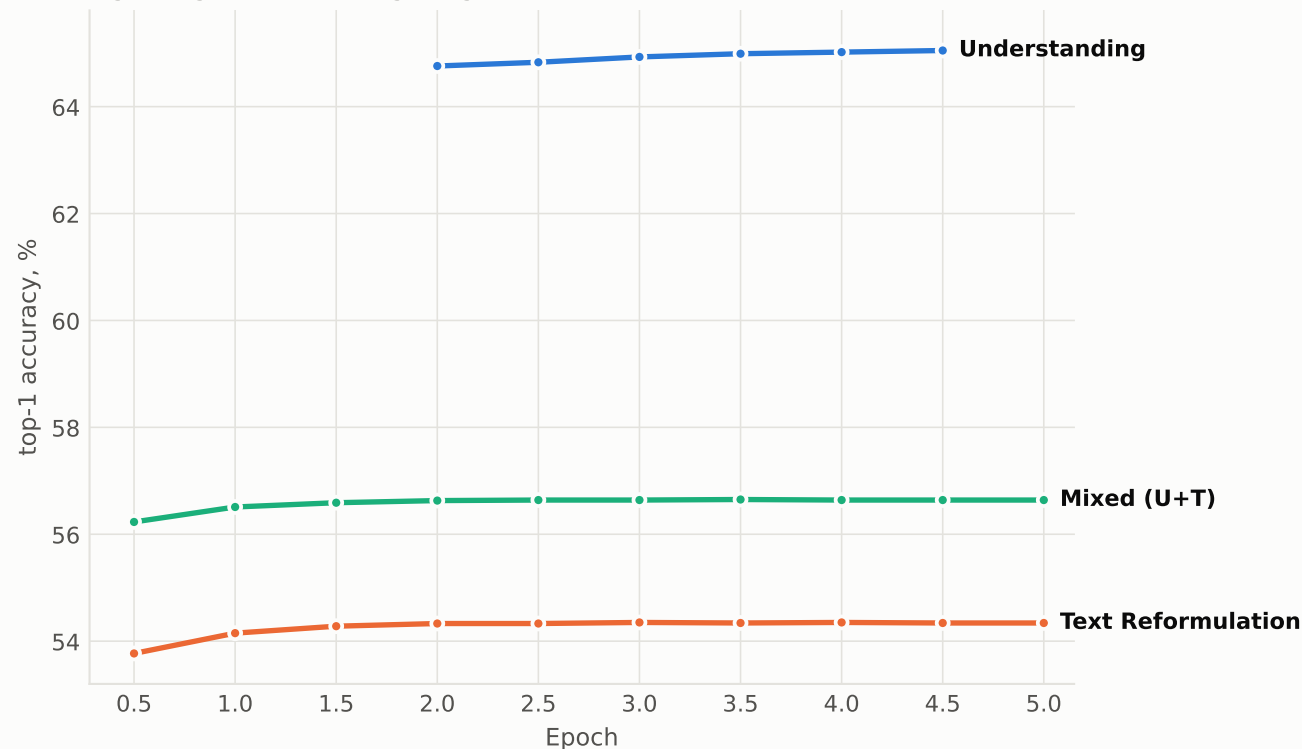
Evaluation loss

$0.5 \cdot \text{CE} + 0.5 \cdot \text{KL}$, lower is better



Top-1 accuracy

argmax agreement with target, higher is better



Understanding was launched for 25 epochs and stopped early at epoch 4.5 on plateau; its curve therefore starts at the first retained checkpoint (epoch 2.0). All three models plateau within 2–3.5 epochs.